

PHASE 2 – REQUIREMENT ANALYSIS

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders

1. Introduction

The Requirement Analysis Phase identifies the requirements needed to develop the automated IT procurement system. The requirements are collected based on the needs of employees, managers, IT teams, procurement teams, and administrators.

2. Stakeholders

The main stakeholders of the system are:

- Employees – Submit laptop requests.
- Managers – Review and approve requests.
- IT Team – Verify technical requirements.
- Procurement Team – Process laptop orders.
- Administrator – Manage users and system information.

3. Functional Requirements

The system should provide the following functions:

1. User login and authentication.
2. Employee registration and profile management.
3. Display available standard laptop options.
4. Submit laptop purchase/request forms.

5. Manager approval or rejection.
6. IT/Procurement review.
7. Update request status.
8. Track the progress of laptop requests.
9. Store request and user information.
10. Provide notifications or workflow updates.

4. Non-Functional Requirements

The system should satisfy the following requirements:

- Usability: Easy and simple user interface.
- Performance: Requests should be processed quickly.
- Security: User information should be protected.
- Reliability: System should work consistently.
- Maintainability: Easy to update and maintain.
- Scalability: System should support additional users and requests.

5. Input Requirements

The system may collect:

- Employee name and ID.
- Department.
- Laptop model/type.
- Required specifications.
- Business justification.
- Request date.
- Approval details.

6. Output Requirements

The system should provide:

- Request confirmation.
- Approval/rejection status.
- Procurement status.
- Order details.
- Request tracking information.
- Completion status.

7. Request Status

The laptop request can follow these statuses:

Submitted → Pending Approval → Approved → IT Review → In Procurement → Ordered
→ Completed

If the request is rejected, the status will be changed to Rejected.

8. Hardware and Software Requirements

Hardware

- Computer/Laptop.
- Minimum 4 GB RAM.
- Internet connection.

Software

- Operating System: Windows/Linux.
- Code Editor: Visual Studio Code.
- Programming technologies as selected for development.
- Database system.
- Web browser.
- Git and GitHub for version control.

9. Business Rules

- Only authorized users can submit or process requests.
- Managers can approve or reject employee requests.
- Procurement processing should happen only after required approvals.
- Every request should have a unique record.
- Request status should be updated at each stage.

10. Expected Outcome

After completing the Requirement Analysis Phase, the team will have a clear understanding of:

- System users and roles.
- Functional requirements.
- Non-functional requirements.
- Required inputs and outputs.
- Approval workflow.
- Hardware and software requirements.

11. Conclusion

The Requirement Analysis Phase defines what the automated IT procurement system should do and what users need from the system. These requirements will be used as the foundation for the Project Design and Development Phases.